Tom Hess

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PRODUCT-FOCUSED, AI-ENABLED ENGINEERING LEADER & SOFTWARE ENGINEER

Software engineer and engineering leader with 15+ years of experience building automation, integrations, and AI-enabled systems for SaaS and enterprise businesses, including two Director-level engineering roles. Hands-on work includes Java gift card and call-routing systems and RPA-driven billing automation. Since 2025, I have shipped apps and built an agent pipeline using an AI coding agent. Three years in product roles added a product lens, so I prioritize engineering work by customer outcome.

TECHNICAL SKILLS

Engineering & Integration: API / ERP / EDI Integrations | SaaS Platform Modernization | Root Cause Analysis | Technical Debt Management | SQL & Data Analysis | Process Mapping & System Diagramming

AI & Automation: Agentic AI Systems Design & Orchestration | AI-Assisted Delivery & Workflow Automation | AI-Assisted Application Development (Python, TypeScript, Rust) | Claude Code

Leadership & Delivery: Team Development & Coaching | Cross-Functional Leadership | Offshore / Distributed Team Management | Vendor Negotiation | Scrum and SAFe practice | Stakeholder Communication

Tools & Platforms: Jira | Confluence | GitLab | Jenkins | AWS | SQL Server | Power BI | Figma

TEAM & LEADERSHIP RECOGNITION

Tech Team of the Year – Cincinnati Business Courier (2019) | MRI Impact Award – Financial Modeling & Workflow Innovation (2018) | Corporate Tech & Innovation Award – Enterprise Automation (2017) | Realcomm Digie Award – Automation Leadership (2017)

EXPERIENCE

Raptor Technologies, Cincinnati, OH
Senior Product Owner, October 2023 – October 2026
(Acquired PublicSchoolWORKS, May 2024; Senior Product Manager until Oct 2025, then Senior Product Owner)

- Championed AI-DLC adoption, replacing manual Scrum story creation with a structured Intent-Elaborate-Build workflow, defining work up front to cut rework and shortening delivery 30% on the first projects.
- Established team AI governance: approved AI tools and MCP servers, AI-specific logins for data protection, and a human-review merge gate for AI-generated code; created and shared 4 AI skills with the team.
- Led product development for compliance-driven LMS and workflow automation platforms serving ~45,000 schools nationwide, with ~99.9% customer retention.
- Eliminated regressions within 90 days through structured peer review, QA pipelines, and domain ownership, reducing support tickets by 20–30% and generating unsolicited positive customer feedback.
- Oversaw two delivery teams post-acquisition: an internal team of 3 engineers, 3 QA, and 1 DevOps lead, plus a vendor partner team of 1 PM, 3 engineers, and shared QA.
- Delivered Vehicle Accident Management, a new product, working closely with customers through definition and an immediate pilot with 3 customers; it drew immediate sales after reaching production.
- Coordinated phased Raptor–PSW SSO integration across three teams, delivering flat-file near-term value while laying groundwork for real-time API-based single sign-on.

Octus (formerly Reorg), Cincinnati, OH
Director of Engineering, July 2022 – September 2023

- Led root cause analysis on database and scalability outages, meeting 24–48 hour resolution SLAs on major incidents, where SLAs had previously been rarely met.
- Directed three cross-functional delivery teams with $500K budget oversight: 10 direct engineering reports plus two additional teams managed through an engineering manager.

- Implemented Scrum practices and documentation standards (runbooks, design docs, onboarding guides), reducing engineer onboarding time by 50%.
- Interviewed and hired engineers, and ran formal performance reviews for 10 direct reports.

Cengage Group, Cincinnati, OH
Manager, Software Engineering, January 2020 – July 2022

- Led distributed onshore and offshore engineering teams (10–12 engineers) supporting Cengage Unlimited, the flagship subscription learning platform serving millions of students.
- Delivered a RESTful Authorization & Permissions API (first touchpoint for every login) with sub-millisecond latency and 99.9% uptime by consolidating 5 legacy services.
- Retired 3 legacy applications through an SOA modernization project, collaborating with 20+ system owners to reduce complexity, maintenance burden, and costs.
- Increased sprint velocity by 50% through SAFe Agile practices, cross-team alignment, and coaching engineers into expanded roles.
- Reduced vendor costs from $40K to $1.2K annually through geolocation vendor consolidation, saving ~$39K a year across teams.

Phillips Edison & Company, Cincinnati, OH
Director / Associate VP, Enterprise Applications, October 2013 – January 2020
(Progressed from Application Developer → Manager → Director → Associate VP over 6+ years)

- Built Tenant Billing Automation Platform integrating ERP, vendor APIs (Billtrust), and RPA, cutting a 2-week billing cycle by 50–60% and freeing billing staff for direct tenant service and other higher-value work.
- Developed the Lease Profitability App alongside Finance and Leasing, cutting deal analysis from hours to seconds for 30 leasing agents nationwide with real-time deal evaluation.
- Grew team from two developers to 12 engineers, contractors, and interns, training them on MRI, a real estate ERP, while overseeing a ~$1M annual budget.
- Built ICSC conference kiosk application delivering live leasing and availability data to agents during tenant negotiations, reducing deal close time by 1–2 days.
- Wrote and ran SQL scripts and built stored procedures and dashboards in the ERP and Excel cubes, supporting about 10 Finance and Leasing staff (5 each) with the data behind their decisions.
- Built a SQL Server training plan from Microsoft certification coursework into each engineer's growth path, reaching 90% completion and a ~10% YoY reduction in support bugs over 4 years.

Sterling Jewelers, Fairlawn, OH
Senior Programming Analyst, November 2010 – October 2013

- Built a nationwide gift card processing application in Java for Sterling Jewelers' 13 brands, eliminating external vendor reliance and bringing processing fully in-house.
- Built an automated IVR/call tree system in Java with real-time API integrations, reducing average customer call handling time by 25%.
- Unified the call-center agent desktop across CRM, external payment vendors, and credit account systems with IVR-triggered automation, reducing manual system-switching for agents supporting 13 retail brands.

Barnes Distribution, Cleveland, OH
Programmer Analyst II → III, January 2009 – November 2010

- Led accounts payable side of the EDI payments project, automating JD Edwards data-file download and import steps and cutting check printing from 2 days to a couple of hours.
- Built VB.NET desktop and handheld-device applications for sales agents servicing clients, with the two communicating for ordering and custom reporting.

KeyBank, Cleveland, OH
Application Systems Programmer, July 2007 – December 2008

- Developed and supported high-volume financial transaction systems and SOAP/XML APIs for ATM and banking operations on a team of about 10 engineers.

The Ohio State University, Columbus, OH
Research & Teaching Assistant, September 2005 – June 2007

- Completed 2 years of M.S. research at the Center for Automotive Research on simulating a hybrid vehicle from software components.
- Taught introductory Java programming and software component modeling for two academic years.

SELECTED ENGINEERING PROJECTS

Independent Software Projects
Personal AI & Application Development, 2025 – Present
(Personal projects, built with an AI-enabled coding agent • Crosscue • CableHop • Retirement Planner)

- Built a 7-stage, local-first LLM agent pipeline with phase gates, then retired it for a leaner hosted harness when usage data showed guarded review wasn't worth maintaining.
- Shipped Crosscue, a Flutter crossword app, to 2 app stores (iOS and Google Play), with automated daily puzzle retrieval and an account-free leaderboard on Cloudflare Workers and D1.
- Shipped CableHop through 2 channels (Mac App Store and notarized direct download), a Swift macOS app with a custom C USB/MTP engine for managing Android devices over MTP.
- Rewrote a vanilla JS retirement planner as a React app with 98 automated tests, running the Monte Carlo simulation in a Web Worker, with Zod-validated forms and reducer-based state.

EDUCATION

M.S.
Computer Science & Engineering
The Ohio State University
Thesis: Simulation of a Hybrid Vehicle Based on Software Components

B.S.
Computer Science & Engineering
The Ohio State University

CERTIFICATIONS

ITIL Foundation Certificate | Certified BMC Remedy Administrator | Claude Code 101 | Google Cloud Agent Fundamentals

TECHNICAL COACHING & APPLIED EXPERIENCE

FIRST Tech Challenge (FTC)
Robotics Head Coach – Software Development, 2023 – Present

- Coached teams to two World Championship qualifications (2023–24, 2024–25), including a top-4 finish at the Ohio State Championship and the software-specific FTC Control Award in 2024–25.
- Taught the student team software engineering over 4 seasons: methods and GitHub collaboration, then layered robot code built up from motor and sensor classes, now AI-DLC with human oversight.

COMMUNITY & VOLUNTEERING

Crayons to Computers – Supply Distribution for Underserved Students | Hamilton County Parks – Seasonal Cleanup & Restoration Volunteer